

Program : Diploma in Polymer Technology	
Course Code : 4099	Course Title: Rubber Products Workshop
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To impart skills on manufacturing of moulded rubber products.

Course Prerequisites & Corequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I & II	1, 2
Rubber compounding, moulding and curing		Technology of elastomers	3
Rubber mastication and compounding		Rubber technology workshop	3
Mixing and Processing machinery, two roll mill, extruder, calendar, hydraulic press.		Polymer processing and machinery	4

Course Outcomes:

On completion of the course ,the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Demonstrate the manufacture of moulded rubber products.	23	Applying
CO2	Demonstrate expanded, hard rubber, extruded and calendered products.	20	Applying
	Lab Exam	2	

CO-PO mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3	3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the manufacture of moulded rubber products.		
M1.01	Practice the production of Table mat.	3	Applying
M1.02	Practice the production of Door mat (coloured/black)	3	Applying
M1.03	Practice the production of coloured Paper weight	3	Applying
M1.04	Practice the production of I.B caps	3	Applying
M1.05	Practice the production of Eraser.	3	Applying
M1.06	Practice the production of Gasket.	3	Applying
M1.07	Practice the production of V strap	3	Applying
M1.08	Practice the production of Play ball	2	Applying
	Lab exam I	1	
CO2	Demonstrate the manufacture of expanded, hard rubber, extruded and calendered products.		
M2.01	Practice the production of Hawaii sole.	3	Applying
M2.02	Practice the production of MC	3	Applying
M2.03	Practice the production of Sponge	3	Applying
M2.04	Practice the production of ebonite	3	Applying
M2.05.	Practice the production of any extruded product (tread, channel, hose)	3	Applying
M2.06.	Practice the production of calendered sheet	3	Applying
M2.07.	Practice the production of reinforced hand-made assembled hose.	2	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	Rubber Technology And Manufacture- C.M.Blow (Publisher – Butterworth Hainemann Ltd, 1982 Edition)
R2	Rubber Technology -Maurice Morton (Publisher – Springer, 1999 Edition)
R3	Vanderbilt Rubber Hand book -R.T Vanderbilt (Publisher – R.T Vanderbilt, 1990 Edition)
R4	Training Manual - RTI, RRII, Kottayam
R5	Rubbery Materials and their compounds - J.A Brydson (Publisher –Elsevier Applied Science, London, 1988)

Online Resources:

Sl.No	Website Link
1	www.researchgate.net
2	www.scribd.com
3	www.iri.net.in
4	www.sciencedirect.com